Regession Assignment

1.Problem Statement:

- Stage1: Machine learning
- Stage2: Supervised learning
- Stage3: Regression

2.Rows & Columns:

- Total Number of columns: 5 columns
- Total Number of rows: 1338 rows

3.Pre_processing Method:

- dataset=pd.get_dummies(dataset,dtype=int,drop_first=True)
- Given input from the database two datas are string type by using “get_dummies” function convert it into the numerical type.
- To avoid repeated information “drop_first=True” function to drop first columns.

4.Final Good model with R2 value:

- Multiple linear regression R2_score= 0.78
- Support Vector regressor R2_score= 0.856
- Decision tree R2_score = 0.733
- Random forest R2_score= 0.854

Here final good model is Support vector regressor.

Support vector Machine (default “rbf”)

C	**Linear**	**rbf**	**Poly**	**sigmoid**
10	0.462	-0.032	0.038	0.039
100	0.628	0.320	0.617	0.527
500	0.763	0.664	0.826	0.444
1000	0.764	0.810	0.856	0.287

Decision Tree (default “squared_error” and “best”)

Criterion	**Splitter**	**R2_score**
Squared error	Best	0.687
Squared error	Randon	0.72
Friedman.mse	Best	0.683
Friedman.mse	Random	0.715
Absolute error	Best	0.683
Absolute error	Random	0.733
Poisson	Best	0.722
poisson	random	0.707

Random Forest (default “squared_error” and “100”)

N_estimators	**Criterion**	**R2_score**
50	Squared error	0.849
100	Squared error	0.853
500	Squared error	0.853
50	Absolute error	0.852
100	Absolute error	0.853
500	Absolute error	0.853
50	Poisson	0.849
100	Poisson	0.852
500	Poisson	0.853
50	Friedman.mse	0.850
100	Friedman.mse	0.854
500	Friedman.mse	0.852

Here is the final result.

While comparing all the models the R2_score high in **Support vector Regressor (“poly”, C=1000) is 0.856**